Assignment 5: - Graph Analytics: University Course Registration System Using Neo4j

Problem statement

A university wants to develop a graph database to manage students, faculty members, courses, and academic departments. The system should represent these entities as nodes and their associations as relationships. You are required to implement the system using Neo4j and the Cypher query language. The sample data and relationships are provided below. Your task is to write the Cypher code to create the nodes, assign properties, establish relationships, and answer the given questions through queries.

1. Sample data

A. Student

Student ID	Name	Semester	Email
S101	Aarav Sharma	5	aarav@univ.edu
S102	Priya Das	5	priya@univ.edu
S103	Rahul Verma	3	rahul@univ.edu
S104	Sneha Roy	3	sneha@univ.edu
S105	Arjun Patel	5	arjun@univ.edu

B. Faculty

Faculty ID	Name	Designation	Specialization
F101	Dr. Meera Sen	Associate Professor	Database Systems
F102	Dr. Amit Roy	Assistant Professor	Data Analytics

C. Courses

Course Code	Course Title	Credits	Level
CS301	Graph Analytics	3	Advanced
CS302	Database Management Systems	4	Core
CS303	Data Mining	3	Advanced
CS304	Big Data Analytics	4	Advanced
CS305	NoSQL Databases	3	Core

D. Departments

Department ID	Department Name	Building
D101	Computer Science	Block A

D102	Information Technology	Block B

2. Relationships to create

Create the following relationships with the specified direction.

Relationship	Direction	Meaning
ENROLLED_IN	Student → Course	Student enrolment in a course
TEACHES	Faculty → Course	Faculty member teaching a course
BELONGS_TO	Student → Department	Student's department
OFFERS	Department → Course	Department offering a course

A. Student enrolments

Student	Enrolled courses
Aarav Sharma	Graph Analytics, DBMS, NoSQL Databases
Priya Das	Graph Analytics, Data Mining
Rahul Verma	DBMS, Big Data Analytics
Sneha Roy	Data Mining, NoSQL Databases
Arjun Patel	Graph Analytics, Big Data Analytics

B. Faculty teaching assignments

Faculty	Courses taught
Dr. Meera Sen	DBMS, NoSQL Databases
Dr. Amit Roy	Graph Analytics, Data Mining, Big Data Analytics

C. Student–department associations

Students	Department
Aarav Sharma, Priya Das, Rahul Verma	Computer Science
Sneha Roy, Arjun Patel	Information Technology

D. Department–course associations

Department	Courses offered
Computer Science	DBMS, Data Mining, NoSQL Databases
Information Technology	Graph Analytics, Big Data Analytics

3. Properties of relationships

Assign the following properties to the relevant relationships.

- ENROLLED_IN: semesterEnrolled, grade
- TEACHES: academicYear

Use suitable values for these properties. For example, semesterEnrolled can be 3 or 5, grade can be Not Published, and academicYear can be 2026-27.

4. Programming tasks

Using Neo4j Browser or Neo4j Desktop, write Cypher queries to perform the following operations.

Part A — Graph creation and verification

1. Create all student, faculty, course, and department nodes with the given properties.
2. Create all the specified relationships with their appropriate directions.
3. Assign properties to the ENROLLED_IN and TEACHES relationships.
4. Display the complete graph.
5. Count the number of nodes belonging to each label.
6. Display all relationship types and verify their directions.

Part B — Retrieval and filtering queries

7. Retrieve all students studying in Semester 5.
8. Find all courses with more than 3 credits.
9. Retrieve the courses taught by Dr. Meera Sen.
10. Find all students enrolled in Graph Analytics.
11. Display the names of all students belonging to the Computer Science department.
12. Find all courses offered by the Information Technology department.
13. Retrieve all students enrolled in NoSQL Databases along with their semester and grade.
14. Find all students enrolled in both Graph Analytics and another course.

Submission Guidelines:

a) Use the .docx file previously shared for assignment submission

b) Put your name reg no and other details

c) Put the problem statement (The above text)

d) Keep all d1) Cypher queries written for the tasks d2) the output along with the d3) Neo4j screenshots for the steps mentioned in Part A and Part B.